Merged Data, DAX, and DAX Function Insights

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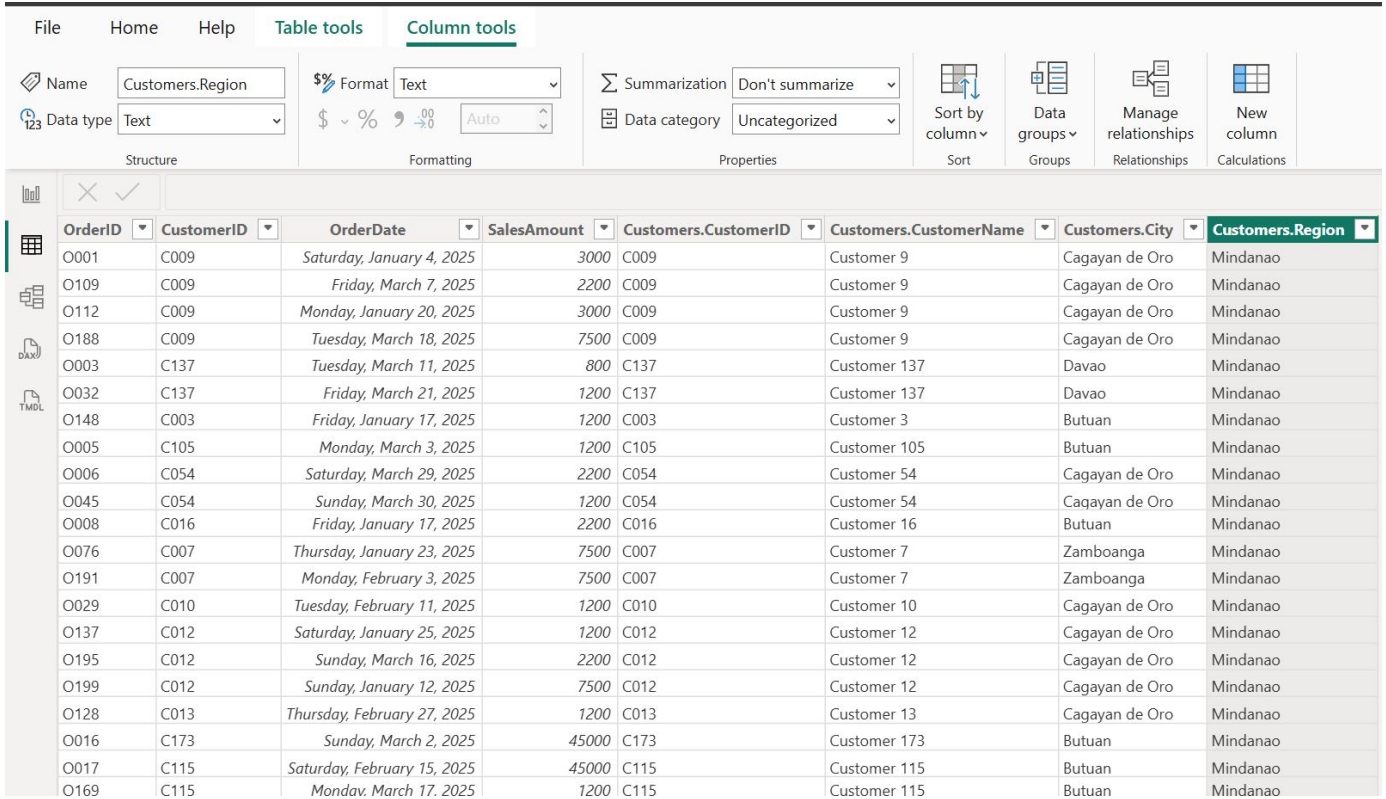
IS 107 - Business Intelligence

Part 1: Review the Merged Dataset



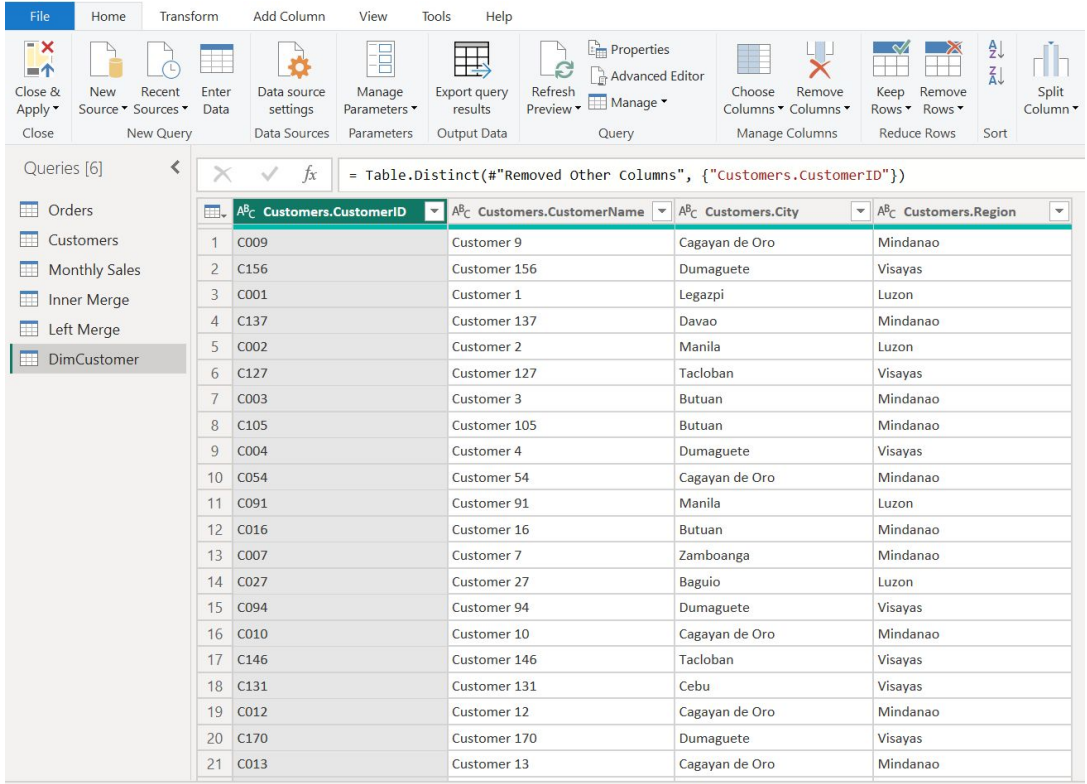
OrderID	CustomerID	OrderDate	SalesAmount	Customers.CustomerID	Customers.CustomerName	Customers.City	Customers.Region
O001	C009	Saturday, January 4, 2025	3000	C009	Customer 9	Cagayan de Oro	Mindanao
O109	C009	Friday, March 7, 2025	2200	C009	Customer 9	Cagayan de Oro	Mindanao
O112	C009	Monday, January 20, 2025	3000	C009	Customer 9	Cagayan de Oro	Mindanao
O188	C009	Tuesday, March 18, 2025	7500	C009	Customer 9	Cagayan de Oro	Mindanao
O003	C137	Tuesday, March 11, 2025	800	C137	Customer 137	Davao	Mindanao
O032	C137	Friday, March 21, 2025	1200	C137	Customer 137	Davao	Mindanao
O148	C003	Friday, January 17, 2025	1200	C003	Customer 3	Butuan	Mindanao
O005	C105	Monday, March 3, 2025	1200	C105	Customer 105	Butuan	Mindanao
O006	C054	Saturday, March 29, 2025	2200	C054	Customer 54	Cagayan de Oro	Mindanao
O045	C054	Sunday, March 30, 2025	1200	C054	Customer 54	Cagayan de Oro	Mindanao
O008	C016	Friday, January 17, 2025	2200	C016	Customer 16	Butuan	Mindanao
O076	C007	Thursday, January 23, 2025	7500	C007	Customer 7	Zamboanga	Mindanao
O191	C007	Monday, February 3, 2025	7500	C007	Customer 7	Zamboanga	Mindanao
O029	C010	Tuesday, February 11, 2025	1200	C010	Customer 10	Cagayan de Oro	Mindanao
O137	C012	Saturday, January 25, 2025	1200	C012	Customer 12	Cagayan de Oro	Mindanao
O195	C012	Sunday, March 16, 2025	2200	C012	Customer 12	Cagayan de Oro	Mindanao
O199	C012	Sunday, January 12, 2025	7500	C012	Customer 12	Cagayan de Oro	Mindanao
O128	C013	Thursday, February 27, 2025	1200	C013	Customer 13	Cagayan de Oro	Mindanao
O016	C173	Sunday, March 2, 2025	45000	C173	Customer 173	Butuan	Mindanao
O017	C115	Saturday, February 15, 2025	45000	C115	Customer 115	Butuan	Mindanao
O169	C115	Monday, March 17, 2025	1200	C115	Customer 115	Butuan	Mindanao

Examine the existing columns from previous LW

Part 1: Review the Merged Dataset | Guide Questions

1. Is the dataset in flat-table format?
   - **Yes, the dataset is in a flat-table format because all the data is stored in a single table with rows and columns.**
2. Which columns describe customers?
   - **Customer columns** include **Customers.CustomerID, Customers.CustomerName, Customers.City, and Customers.Region.**
3. Which columns represent transactions?
   - Transaction columns include **OrderID, CustomerID, OrderDate, and SalesAmount**.

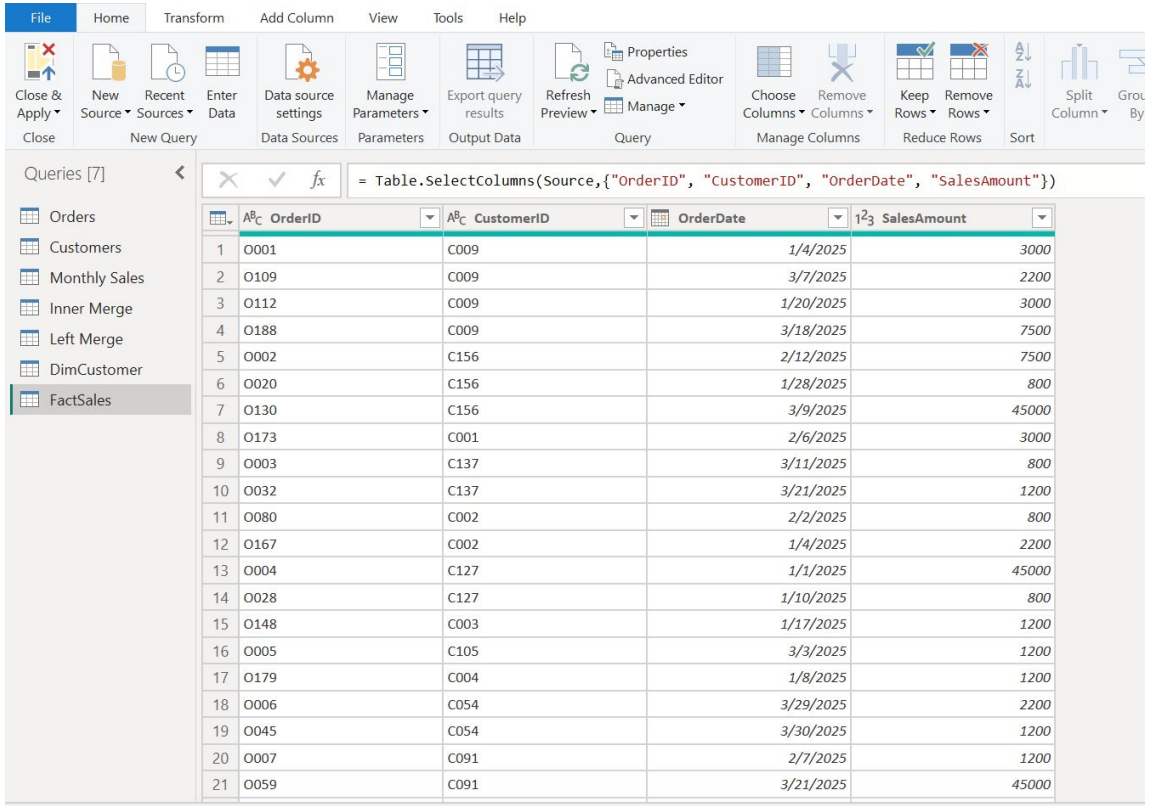
Part 2: Identify Fact and Dimension Tables | DimCustomer



	Customers.CustomerID	Customers.CustomerName	Customers.City	Customers.Region
1	C009	Customer 9	Cagayan de Oro	Mindanao
2	C156	Customer 156	Dumaguete	Visayas
3	C001	Customer 1	Legazpi	Luzon
4	C137	Customer 137	Davao	Mindanao
5	C002	Customer 2	Manila	Luzon
6	C127	Customer 127	Tacloban	Visayas
7	C003	Customer 3	Butuan	Mindanao
8	C105	Customer 105	Butuan	Mindanao
9	C004	Customer 4	Dumaguete	Visayas
10	C054	Customer 54	Cagayan de Oro	Mindanao
11	C091	Customer 91	Manila	Luzon
12	C016	Customer 16	Butuan	Mindanao
13	C007	Customer 7	Zamboanga	Mindanao
14	C027	Customer 27	Baguio	Luzon
15	C094	Customer 94	Dumaguete	Visayas
16	C010	Customer 10	Cagayan de Oro	Mindanao
17	C146	Customer 146	Tacloban	Visayas
18	C131	Customer 131	Cebu	Visayas
19	C012	Customer 12	Cagayan de Oro	Mindanao
20	C170	Customer 170	Dumaguete	Visayas
21	C013	Customer 13	Cagayan de Oro	Mindanao

Creating DimCustomer by duplicating the previous merged table and removing the duplicate ID

Part 2: Identify Fact and Dimension Tables | Create FactSales



	OrderID	CustomerID	OrderDate	SalesAmount
1	O001	C009	1/4/2025	3000
2	O109	C009	3/7/2025	2200
3	O112	C009	1/20/2025	3000
4	O188	C009	3/18/2025	7500
5	O002	C156	2/12/2025	7500
6	O020	C156	1/28/2025	800
7	O130	C156	3/9/2025	45000
8	O173	C001	2/6/2025	3000
9	O003	C137	3/11/2025	800
10	O032	C137	3/21/2025	1200
11	O080	C002	2/2/2025	800
12	O167	C002	1/4/2025	2200
13	O004	C127	1/1/2025	45000
14	O028	C127	1/10/2025	800
15	O148	C003	1/17/2025	1200
16	O005	C105	3/3/2025	1200
17	O179	C004	1/8/2025	1200
18	O006	C054	3/29/2025	2200
19	O045	C054	3/30/2025	1200
20	O007	C091	2/7/2025	1200
21	O059	C091	3/21/2025	45000

Creating FactSales by duplicating the previous merged table and removing the duplicate ID

Part 2: Identify Fact and Dimension Tables | Guide Questions

1. Why must dimension tables contain unique keys?

- Dimension tables must contain unique keys so each record represents only one entity, such as a single customer. This helps create accurate relationships with fact tables and prevents incorrect data matching.

2. What problems occur if duplicates exist in DimCustomer?

- Duplicates in **DimCustomer** can cause incorrect relationships, duplicate results in reports, and inaccurate calculations when analyzing sales data.

Part 3: Define Primary & Foreign Keys | Guide Questions

Identify Keys

Table	Key	Type
DimCustomer	CustomerID	Primary Key
FactSales	CustomerID	Foreign Key
FactSales	OrderID	Primary Key

	Customers.CustomerID	Customers.CustomerName	Customers.City	Customers.Region
1	C009	Customer 9	Cagayan de Oro	Mindanao
2	C009	Customer 9	Cagayan de Oro	Mindanao

	OrderID	CustomerID	OrderDate	SalesAmount
1	O001	C009	1/4/2025	3000

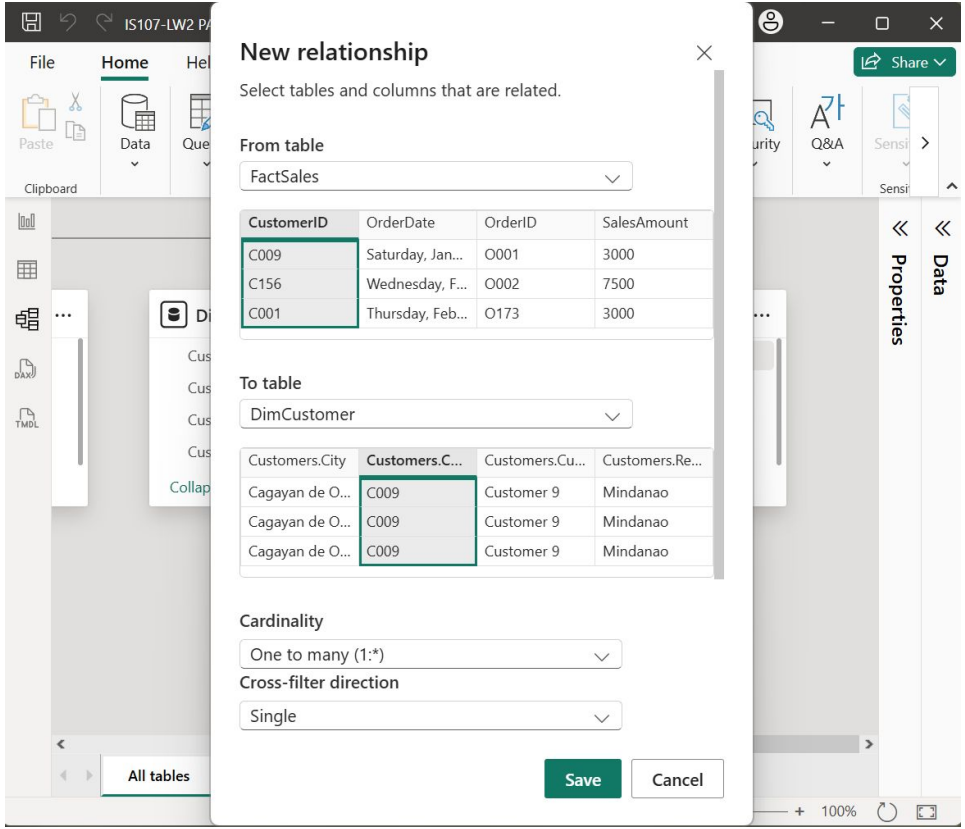
1. What is a primary key?

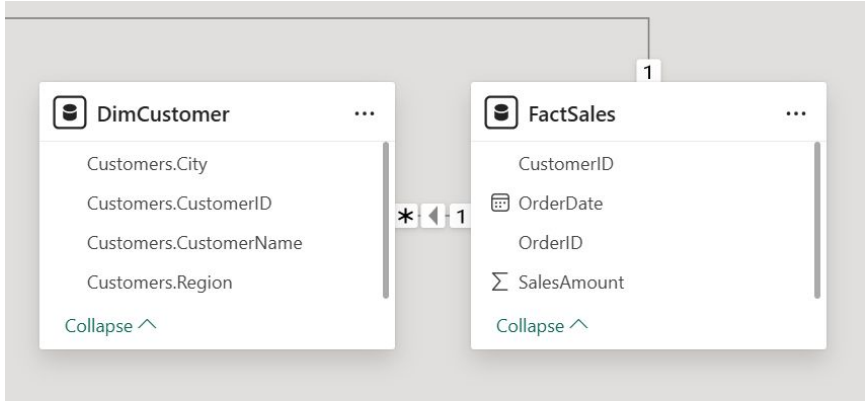
- A **primary key** is a column that uniquely identifies each record in a table, ensuring no duplicate values exist.

2. Why is CustomerID a foreign key in FactSales?

- **CustomerID** is a foreign key in **FactSales** because it links each sales transaction to the corresponding customer in **DimCustomer**.

Part 4: Create Relationships





Creating relationship between DimCustomer and FactSales by dragging CustomerID of FactSales to DimCustomer then changing the Cardinality to one to many.

Part 4: Create Relationships | Guide Questions

1. Why should relationships flow from dimension to fact?

- Relationships should flow from dimension tables to fact tables because dimension tables contain descriptive data, while fact tables store transactions. This structure helps Power BI filter and analyze data correctly.

2. What happens if the relationship is Many-to-Many?

- A **Many-to-Many relationship** can cause incorrect calculations and ambiguous results because multiple rows in both tables may match each other.

Part 5: Validate the Data Model

Customers.CustomerName	Customers.Region	Sum of SalesAmount
Customer 1	Luzon	5200
Customer 10	Mindanao	1200
Customer 100	Mindanao	16500
Customer 101	Mindanao	6000
Customer 102	Luzon	90800
Customer 103	Mindanao	4200
Customer 104	Visayas	3700
Customer 105	Mindanao	1200
Customer 106	Visayas	52500
Customer 107	Mindanao	9000
Customer 108	Mindanao	46200
Customer 11	Visayas	1200
Customer 110	Visayas	47200
Customer 111	Mindanao	800
Customer 112	Mindanao	5400
Customer 113	Luzon	91500
Customer 114	Mindanao	5200
Customer 115	Mindanao	46200
Customer 116	Luzon	49200
Customer 118	Mindanao	49500
Total		**3894400**

Validating the data and verifying if the data aggregate correctly by making a table for visual result.

Part 5: Validate the Data Model | Guide Questions

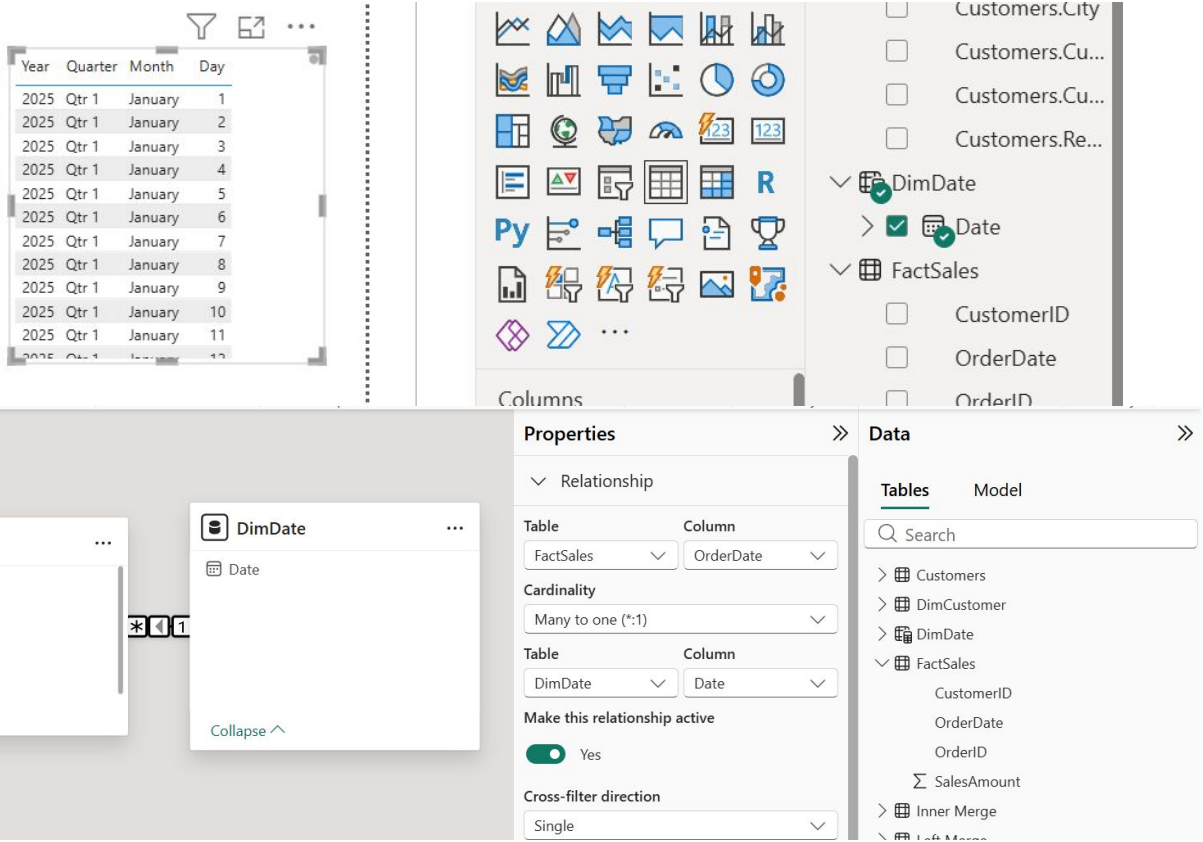
1. Did sales aggregate correctly per customer?

- Yes, sales aggregated correctly because the SalesAmount values were summed based on each customer, showing total sales per customer.

2. What would happen if relationships were missing?

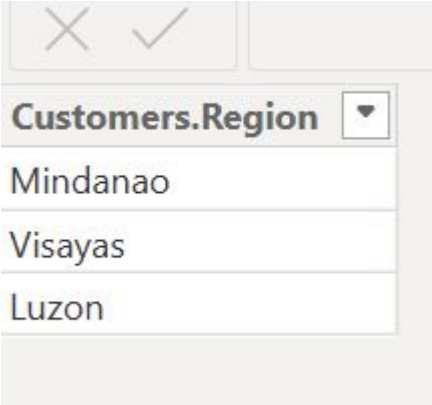
- If relationships were missing, customer data and sales data would not connect, resulting in incorrect or repeated values and inaccurate reports.

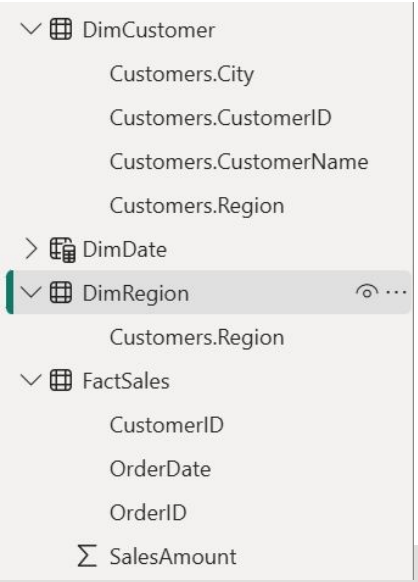
Enhancement 1: Add DimDate Table

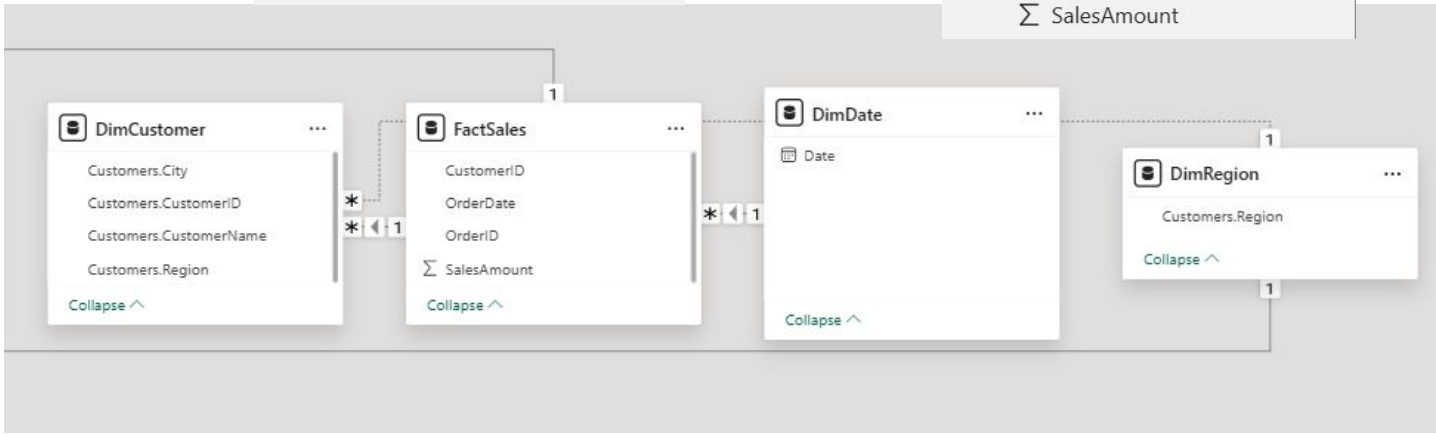


Creating new table by pasting the provided formula and making a relationship between the two table for the Date.

Enhancement 2: Create a DimRegion Table







Creating new Region tablemaking a relationship between the two table for the Region.

Enhancement 3: Compare Flat vs Star Schema Performance

Customers.CustomerName	Customers.Region	Sum of SalesAmount
Customer 1	Luzon	5200
Customer 10	Mindanao	1200
Customer 100	Mindanao	16500
Customer 101	Mindanao	6000
Customer 102	Luzon	90800
Customer 103	Mindanao	4200
Customer 104	Visayas	3700
Customer 105	Mindanao	1200
Customer 106	Visayas	52500
Customer 107	Mindanao	9000
Customer 108	Mindanao	46200
Customer 11	Visayas	1200
Customer 110	Visayas	47200
Customer 111	Mindanao	800
Customer 112	Mindanao	5400
Customer 113	Luzon	91500
Customer 114	Mindanao	5200
Customer 115	Mindanao	46200
Customer 116	Luzon	49200
Customer 118	Mindanao	49500
Total		**3894400**

Flat Tables

Customers.CustomerName	Customers.Region	Sum of SalesAmount	Month
Customer 119	Visayas	741200	February
Customer 12	Mindanao	641700	January
Customer 121	Luzon	641700	January
Customer 122	Mindanao	420700	March
Customer 123	Mindanao	641700	January
Customer 124	Visayas	641700	January
Customer 125	Mindanao	420700	March
Customer 126	Luzon	641700	January
Customer 127	Visayas	641700	January
Customer 128	Mindanao	420700	March
Customer 129	Visayas	641700	January
Customer 13	Mindanao	741200	February
Customer 130	Visayas	641700	January
Customer 131	Visayas	420700	March
Customer 132	Mindanao	420700	March
Customer 134	Visayas	420700	March
Customer 135	Luzon	420700	March
Customer 136	Visayas	741200	February
Customer 137	Mindanao	420700	March
Customer 138	Luzon	641700	January
Total		**1848600**	

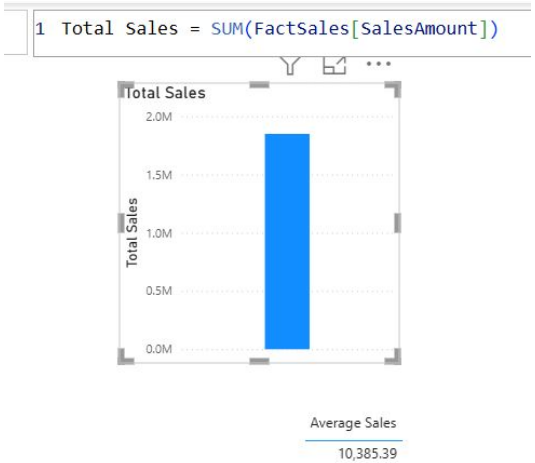
Star Schema

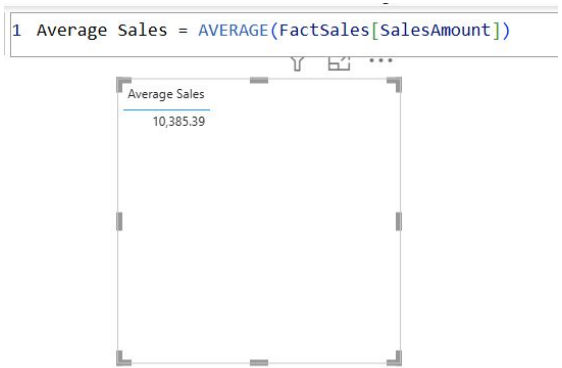
Star schema performs better than a flat table because it separates data into fact and dimension tables, reducing redundancy and improving performance. It provides more accurate aggregations and better filtering, while flat tables can lead to slower performance and duplicate data issues.

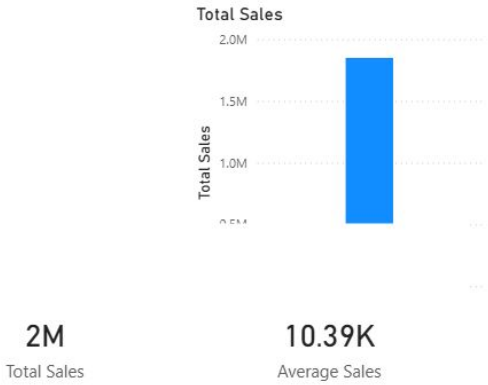
Comparing Flat table and Star Schema by creating two different tables from different data merges.

Introduction to DAX

Introduction to DAX: Create Measures







Creating measurement using DAX prompts. Quantity is not included as it is not in the table.

Introduction to DAX: Create Measures | Guide Questions

1. What is a DAX measure?

- A DAX measure is a calculation created using formulas that dynamically computes values based on the data and filters in a report.

2. Difference between SUM and AVERAGE?

- SUM adds all values in a column, while AVERAGE calculates the mean by dividing the total by the number of values.

3. Why use measures instead of calculated columns?

- Measures are calculated dynamically based on filters and use less memory, while calculated columns store values in the table and increase data size.

Introduction to DAX: Enhancement Activities

KPI Visual with Date as Trend

```
Sales Column = FactSales[SalesAmount]
```

OrderID	CustomerID	OrderDate	SalesAmount	Sales Column
O001	C009	Saturday, January 4, 2025	3000	3000
O002	C156	Wednesday, February 12, 2025	7500	7500
O173	C001	Thursday, February 6, 2025	3000	3000
O003	C137	Tuesday, March 11, 2025	800	800
O080	C002	Sunday, February 2, 2025	800	800
O004	C127	Wednesday, January 1, 2025	45000	45000
O148	C003	Friday, January 17, 2025	1200	1200
O005	C105	Monday, March 3, 2025	1200	1200
O179	C004	Wednesday, January 8, 2025	1200	1200
O006	C054	Saturday, March 29, 2025	2200	2200
O007	C091	Friday, February 7, 2025	1200	1200
O008	C016	Friday, January 17, 2025	2200	2200

Column is stored in memory while Measure is calculated when needed

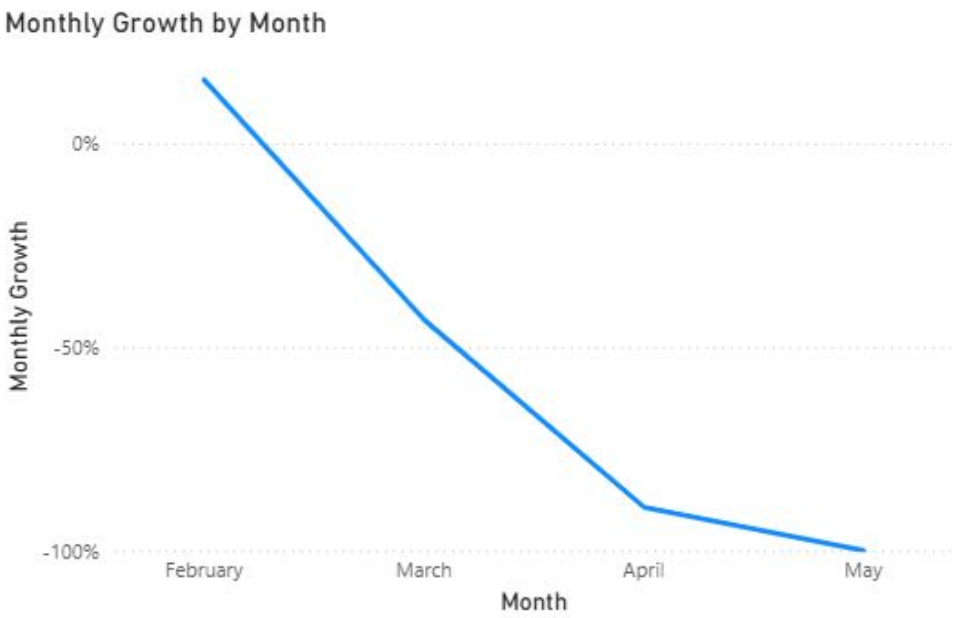
178
Count Sales

178
Row Count

COUNT counts only non-empty values in a specific column, while COUNTROWS counts all rows in a table regardless of missing values.

DAX Functions

DAX Functions



2M

YTD Sales

Line Chart to visualize total YTD Sales per month and a Card for total Sales

DAX Functions | Guide Questions

1. What is time intelligence in DAX?

- Time intelligence in DAX refers to functions that perform calculations based on dates, such as year-to-date or month-to-month comparisons.

2. Why is a date table required?

- A date table is required because time intelligence functions rely on a continuous and structured set of dates to calculate trends correctly.

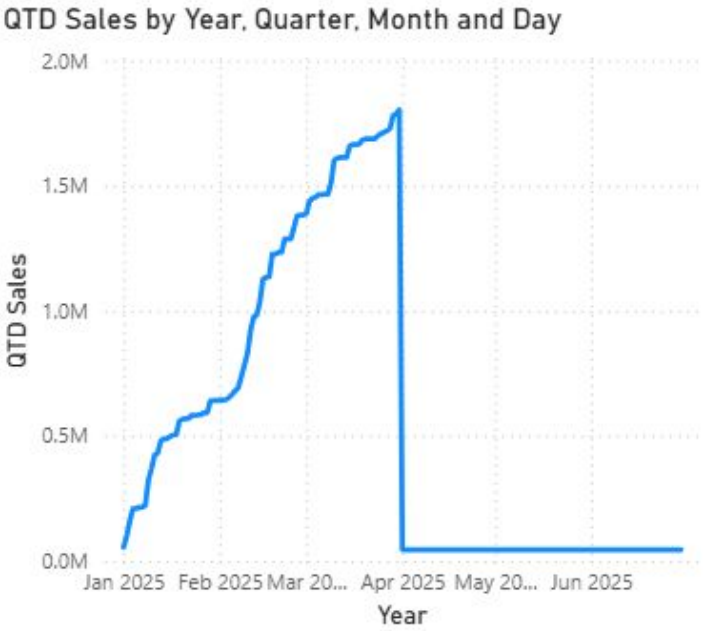
3. How does PREVIOUSMONTH help analyze trends?

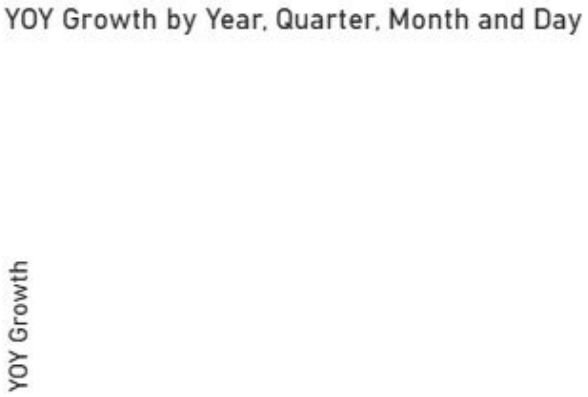
- PREVIOUSMONTH allows comparison of current sales with the previous month, helping identify growth or decline over time.

4. What insights can YTD Sales provide?

- YTD Sales shows total sales from the beginning of the year up to the current date, helping track overall performance and progress.

DAX Functions | Enhancement





QTD Sales shows total sales from the start of the quarter to date. MoM compares sales between consecutive months to track short-term trends. YoY compares sales with the same period last year to show long-term growth.

Note: YoY is blank because there is no previous year data for comparison.